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Lab Rotation Report

Abstract

The hippocampus, a crucial brain region for memory and learning, is vulnerable to aging and vascular pathology. This project investigates the relationship between hippocampal vessel distance, hippocampal volume, and cognitive performance in cognitively normal older adults. Vessel Distance Mapping (VDM) was used to measure the distance from the hippocampus to its nearest blood vessels. The results revealed no significant correlation between vessel distance and cognitive performance. Interestingly, a weak trend indicated that greater vessel distances were associated with larger hippocampal volumes. Furthermore, no significant relationship was found between vessel distance and cardiovascular fitness (VO2 max). These findings suggest that larger hippocampal volumes may be related to greater vessel distances in cognitively normal older adults, which contradicts the initial hypothesis.

Introduction

The hippocampus plays a central role in memory and learning and is highly vulnerable to age-related changes, including vascular insufficiency and structural atrophy. A sufficient blood supply is critical for maintaining hippocampal health, as it supports neuronal metabolism and plasticity(Garcia-Garcia et al., 2023). Variations in hippocampal vascularization may influence cognitive function, particularly in the presence of vascular disease in old age (Garcia-Garcia et al., 2023). Previous studies propose that a closer spatial relationship between blood vessels and hippocampal tissue could help preserve hippocampal volume, which is often associated with

cognitive performance. However, the relationship between hippocampal vessel distance, memory performance, hippocampal volume, and cardiovascular fitness in cognitively normal older adults remains underexplored.

Background

VDM measures the distance between hippocampal tissue and its nearest blood vessels. Garcia-Garcia et al.(2023) suggest that better hippocampal vascularization, indicated by shorter vessel distances, may be linked to preseved cognitive function in older people with cerebral small vessel distance.

Studies have also shown that higher levels of cardiovascular fitness, typically measured by VO2 max, are associated with larger hippocampal volumes and better memory performance. Cardiovascular fitness is thought to enhance cerebral blood flow, thereby improving oxygen delivery to the brain and supporting vascular health (Maass et al., 2015). Aerobic exercise, in particular, has been shown to preserve hippocampal structure and function, potentially mitigating the effects of aging on memory and learning (Maass et al., 2015).

Aims of Study

This study aims to investigate the relationship between hippocampal vessel distance, cognitive performance, and hippocampal volume in cognitively normal older adults. Specifically, it aims to:

1. Quantify hippocampal vessel distance using VDM
2. Analyze the association between vessel distance and cognitive performance
3. Examine the relationship between vessel distance and hippocampal volume.

4. Investigate the relationship between vessel distance and demographic factors (age, sex, education), as well as physical fitness (VO2 max).

Hypotheses:

1. **Cognitive performance:** Shorter vessel distances to the hippocampus are hypothesized to be associated with better cognitive performance.
2. **Hippocampal volume:** Shorter vessel distances are expected to correlate with larger hippocampal volumes..
3. **Physical fitness**: Better cardiovascular fitness (VO2 max) is hypothesized to be associated with shorter vessel distances.

Sample: Data were acquired from 55 older participants (21 females, 34 males) in the CRC1436 B04 project, using 7 Tesla ToF images and hippocampal segmentations. The participants had an average age of 71.07 years. Age, sex, and education levels were included as covariates in analyses. Vessel distance measurements showed a mean of 5.54 mm, a median of 5.57 mm, and a standard deviation of 0.91 mm.

Methods

1. **Preprocessing:**

   Image preprocessing were performed on an T1-weighted images (Figure 1) and ToF images (Figure 2) to obtain skull stripped ToF images for vessel segmentation using the *Omelette* pipeline. Preprocessing was done using MATLAB and SPM12.

   First tissue segmentation was performed on an anatomical MRI image, generating tissue probability maps (Figure 1). From these maps, a brain mask (Figure 3) was created, including only voxels with a combined probability above 0.9. Post-processing was

applied to enhance mask quality by filling holes, removing noise, and smoothing edges. Then, coregistration was performed to align ToF (Figure 2) and T1-weighted images (Figure 1) into a common reference space. The brain mask was applied to both T1 and TOF images, creating skull-stripped versions (Figures 4 and 5).

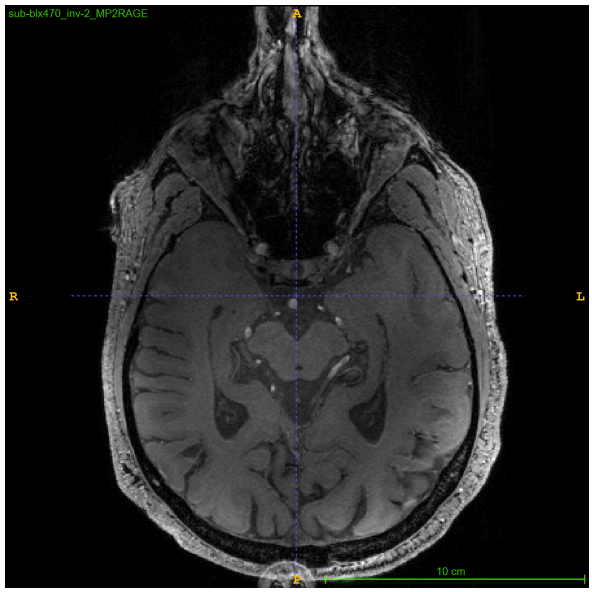

Figure1: T1- weighted image

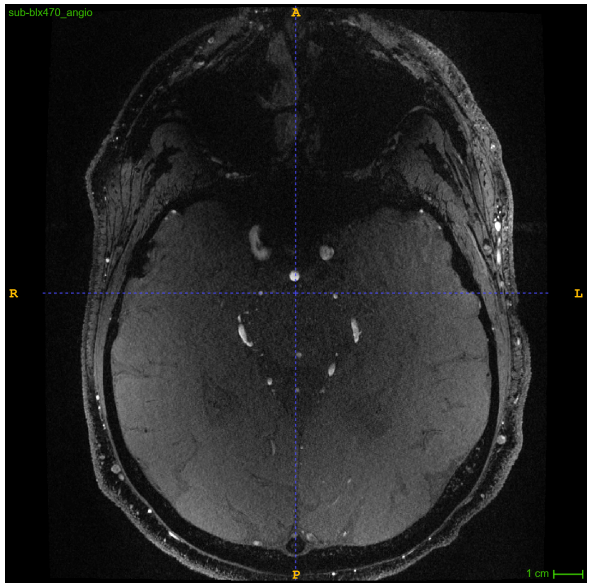

Figure 2: ToF image

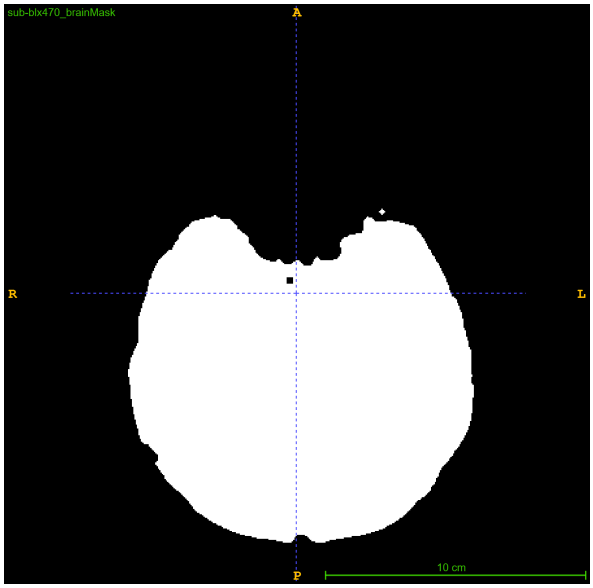

Figure 3: Brain mask

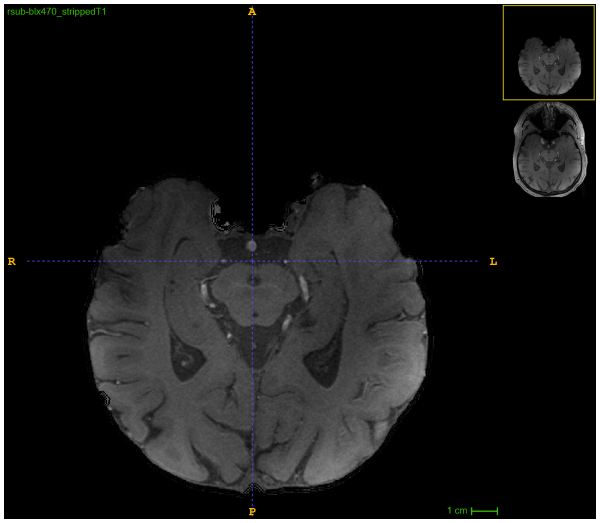

Figure 4: Skull stripped T1-weighted image

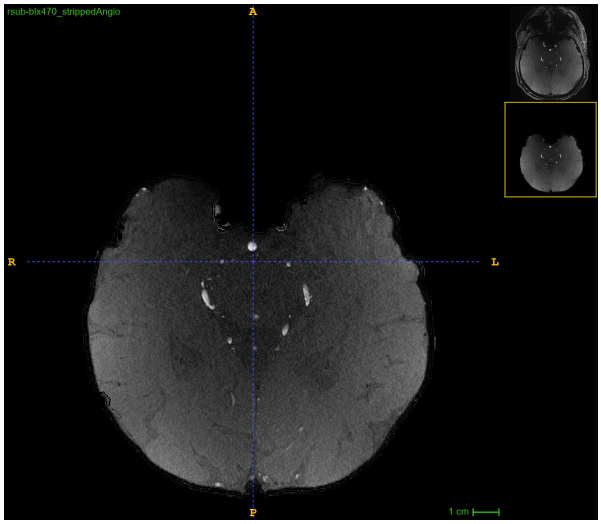

Figure 5: Skull-stripped ToF image

2. **Vessel segmentation:** The preprocessed, skull-stripped ToF images were processed with the small vessel segmentation pipeline "Omelette," written by Hendrik Mattern (Figure 6).

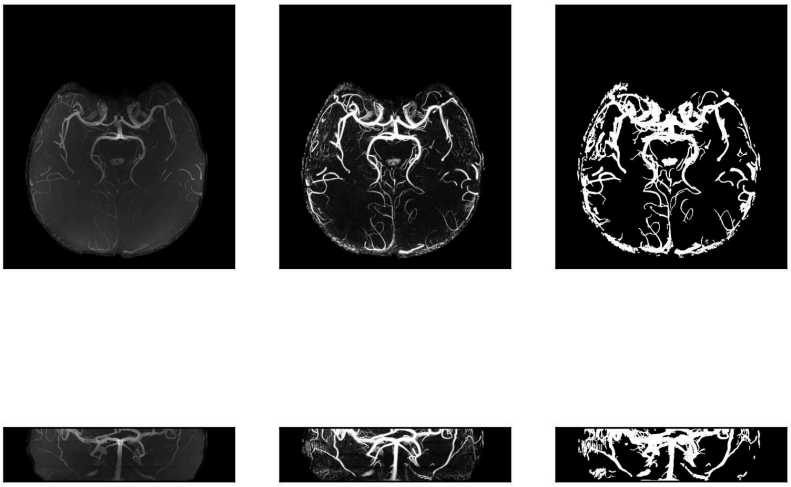

Figure 6: Vessel segmentation of skul-stripped ToF image (tau= 0.5)

3. **Hippocampal Volumetry:** Hippocampal volumes were pre-extracted using the Automated Segmentation of Hippocampal Subfields (ASHS) technique (Figure 7). Quality control was performed on these segmentations. Binary hippocampus masks

(Figure 8) were created from the automatic segmentations. Coregistration of T2-weighted and ToF images to the T1-weighted space was performed to ensure accuracy. Large segmentation errors were corrected, while minor errors were not addressed due to time constraints.

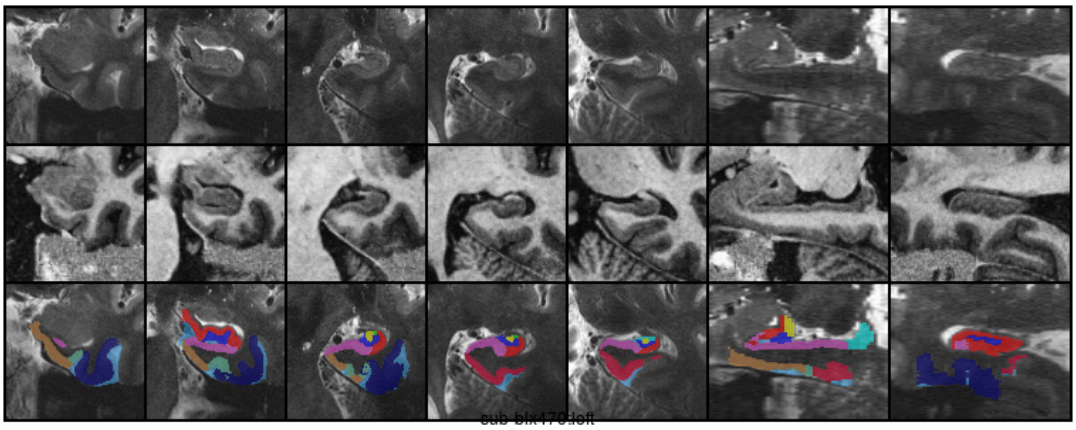

Figure7: ASHS segmentation results

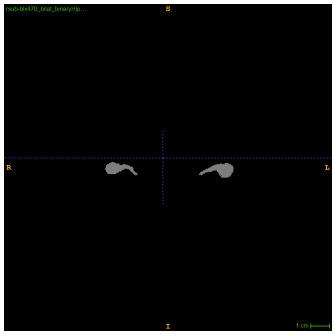

Figure 8: Binary Hippocampus

4. **Vessel Distance Mapping (VDM):** VDM was performed by combining vessel segmentation results with binary hippocampus masks to calculate the distance from each hippocampal voxel to the nearest blood vessel.
5. **Statistical Analysis:** Statistical analyses, including correlation and regression models, were conducted to examine the relationships between hippocampal vessel distance,

cognitive performance, hippocampal volume, and physical fitness. Demographic factors (age, sex, education) were included as covariates in all analyses, except hippocampal volume. Statistical significance was set at $p < 0.05$.

Results

1. **Vessel Distance and Cognitive Performance:** Contrary to the initial hypothesis, verbal memory performance measured via the VLMT (Figures 9 and 10) exhibited a marginally negative trend with vessel proximity ($p= 0.072$). This suggests that closer proximity of blood vessels to the hippocampus is not related to better cognitive function in older adults, specifically verbal memory, as hypothesized . General cognitive performance (Figure 11) measured with the CERAD battery and figural memory (Figure 12) did also not show significant relationships with vessel distance (all $p>0.2$)

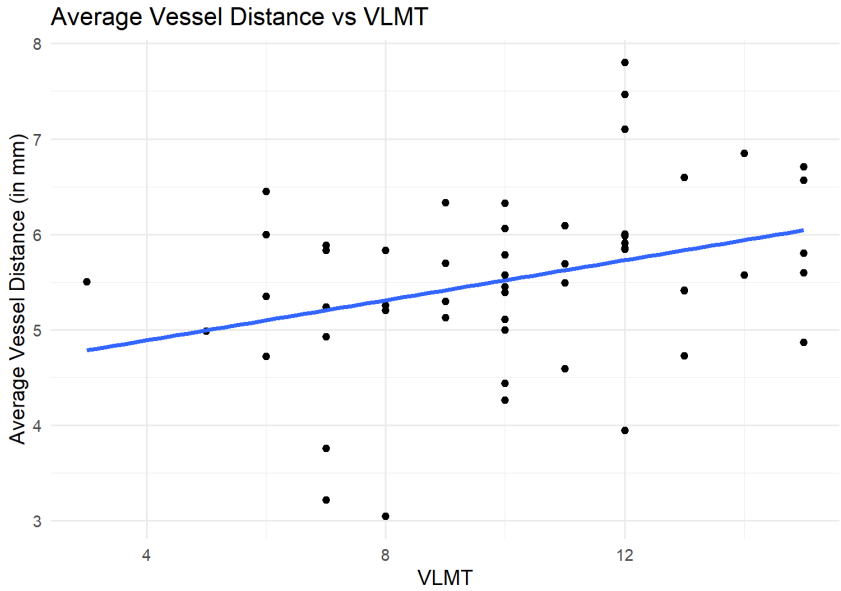

Figure 9: VLMT regression

Characteristic	Beta	95% CI[1]	p-value
avg_vd	0.78	-0.07, 1.6	0.072
age	-0.11	-0.22, 0.01	0.064
sex			
female	—	—	
male	-1.4	-3.1, 0.34	0.11
edu_yrs	0.40	0.05, 0.75	0.028

[1] CI = Confidence Interval

Figure 10: VLMT summary

Characteristic	Beta	95% CI[1]	p-value
avg_vd	0.05	-0.05, 0.15	0.3
age	0.01	0.00, 0.02	0.083
sex			
female	—	—	
male	0.07	-0.13, 0.26	0.5
edu_yrs	0.00	-0.04, 0.04	0.8

[1] CI = Confidence Interval

Figure 11: CERAD summary

Characteristic	Beta	95% CI[1]	p-value
avg_vd	0.43	-1.4, 2.3	0.6
age	-0.34	-0.59, -0.09	0.009
sex			
female	—	—	
male	2.3	-1.5, 6.0	0.2
edu_yrs	0.31	-0.47, 1.1	0.4

[1] CI = Confidence Interval

Figure 12: RCFT summary

2. **Hippocampal Volume**: The relationship between vessel distance and hippocampal volume did not align with the hypothesis that shorter distances would be associated with larger volumes. Instead, the data indicated an unexpected trend, where larger vessel distances were weakly associated with greater hippocampal volumes (Figures 13 and 14, $p < 0.05$).

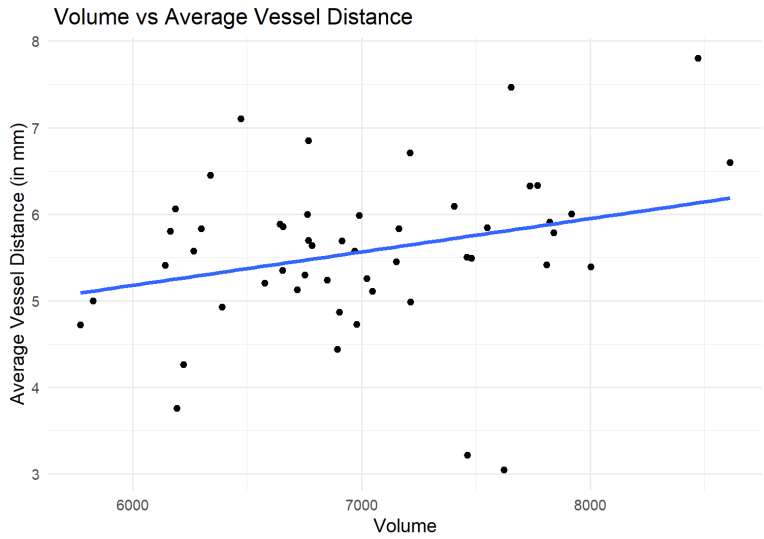

Figure 13: Hippocampal volume regression

Characteristic	Beta	95% CI[1]	p-value
avg_vd	201	1.6, 400	0.048
[1] CI = Confidence Interval			

Figure 14: Hippocampal volume summary

3. **Physical Fitness**:

No significant relationship was found between vessel distance and cardiovascular fitness (VO2 max), suggesting that hippocampal vascularization may not be directly impacted by cardiovascular fitness in old age. However, significant sex differences were observed, with males having higher VO2 max levels overall ($p < 0.05$)(Figure 15). These results

suggest that factors beyond hippocampal vascularization may influence cardiovascular fitness in older adults.

Characteristic	Beta	95% CI[1]	p-value
avg_vd	1.5	-0.67, 3.7	0.2
age	-0.07	-0.34, 0.20	0.6
sex			
female	—	—	
male	4.6	0.59, 8.6	0.026
edu_yrs	0.19	-0.69, 1.1	0.7

[1] CI = Confidence Interval

Figure 15: Fitness summary

Discussion

The results of this study contradict the original hypothesis that shorter vessel distances would correlate with better cognitive performance and larger hippocampal volumes. Instead, vessel proximity was not significantly associated with better performance in cognitively healthy older adults. A potential explanation for these unexpected findings lies in the SPM coregistration process used to align the ASHS hippocampal subfields with structural scans. Coregistration inaccuracies may have introduced spatial misalignment between the hippocampus and vascular maps, leading to erroneous vessel distance measurements. Manual segmentation techniques might provide more precise measurements of hippocampal boundaries and vessel locations.

Furthermore, the unexpected observation that larger hippocampal volumes were weakly associated with greater vessel distances raises important questions about the underlying mechanisms. This finding has been also observed in young adults and might be explained by the

medial location of the vessel relative to hippocampus where larger hippocampi extend more laterally and thereby introduce larger mean distance to the vessels.

References

Maass, A., Berron, D., Libby, L. A., Ranganath, C., & Düzel, E. (2015). Functional subregions of the human entorhinal cortex. *eLife*, *4*. https://doi.org/10.7554/elife.06426

Garcia-Garcia, B., Mattern, H., Vockert, N., Yakupov, R., Schreiber, F., Spallazzi, M., Perosa, V., Haghikia, A., Speck, O., Düzel, E., Maass, A., & Schreiber, S. (2023). Vessel distance mapping: A novel methodology for assessing vascular-induced cognitive resilience. *NeuroImage*, *274*, 120094. https://doi.org/10.1016/j.neuroimage.2023.120094